Final Project Report

Customer Churn Prediction Using Machine Learning

Machine Learning

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1. Introduction

Customer churn occurs when a customer stops using a company service. In the telecommunications sector, churn prediction is an important business problem because retaining existing customers is usually less expensive than acquiring new customers. A churn prediction model can help a company identify customers who are likely to leave and take preventive actions such as offering better plans, discounts, or improved customer support.

The objective of this project is to build a machine learning system that predicts whether a customer will churn based on demographic information, account details, service subscriptions, payment methods, and billing behavior. The project also compares multiple machine learning models and extracts business insights from the dataset.

2. Dataset Description

The project uses the Telco Customer Churn dataset. The dataset contains customer-level information for a telecommunications company, including demographic attributes, account information, subscribed services, charges, and the final churn status.

Feature group	Description
customerID	Unique customer identifier; removed before training because it does not represent customer behavior.
gender, SeniorCitizen, Partner, Dependents	Demographic information about the customer.
tenure	Number of months the customer has stayed with the company.
PhoneService, MultipleLines	Phone-related service information.
InternetService and online services	Internet subscription and additional services such as security, backup, protection, and support.
Contract	Contract type: month-to-month, one year, or two year.
PaperlessBilling, PaymentMethod	Billing and payment behavior.
MonthlyCharges, TotalCharges	Monthly cost and accumulated charges.
Churn	Target variable: Yes means the customer left, No means the customer stayed.

The dataset contains 7,043 records and 21 original columns. The target variable is Churn. The class distribution is imbalanced: 5,174 customers did not churn, while 1,869 customers churned. Therefore, only about 26.54% of customers are churn customers.

Class	Count	Percentage
No Churn	5,174	73.46%
Churn	1,869	26.54%

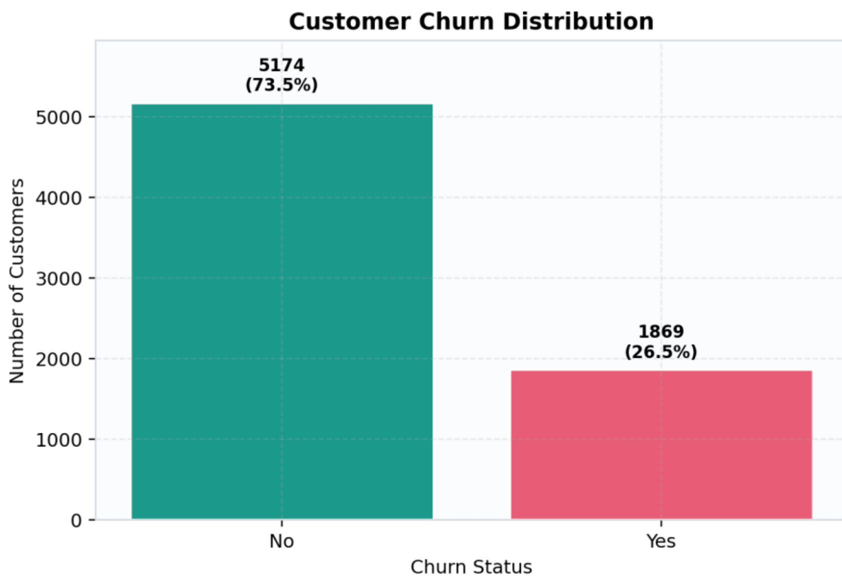

Figure 1. Churn Distribution

Analysis: The figure shows that the dataset is imbalanced. Most customers are in the No Churn class, while the Churn class is smaller. Because of this imbalance, accuracy alone is not enough; recall and F1-score are needed to judge how well the models detect churn customers.

3. Data Preprocessing

The preprocessing stage prepares the raw dataset for machine learning. Since the dataset contains categorical and numerical columns, different transformations are applied to each type. The customerID column is removed because it is only an identifier and does not represent customer behavior.

- The target variable Churn is converted into a binary label: Yes = 1 and No = 0.
- TotalCharges is converted from text to numeric values because it represents a financial value.
- Missing numerical values are filled using the median strategy.
- Missing categorical values are filled using the most frequent category.
- Categorical features are converted into numerical columns using One-Hot Encoding.
- Numerical features are scaled using StandardScaler, which is especially important for SVM.
- The preprocessing pipeline is implemented using scikit-learn ColumnTransformer and Pipeline to avoid data leakage.

Although the raw summary shows zero missing values, TotalCharges is stored as text and includes blank values. After converting it to numeric, those blank values become missing values. The implemented preprocessing pipeline correctly handles this issue using imputation.

Data component	Processing step	Reason
customerID	Dropped	Identifier only; not useful for prediction.
Churn	Mapped to 0/1	Required for classification.
TotalCharges	Converted to numeric	Allows mathematical processing and imputation.
Numerical features	Median imputation + StandardScaler	Handles missing values and scales magnitudes.
Categorical features	Most-frequent imputation + OneHotEncoder	Transforms text categories into numeric model inputs.

4. Sampling Strategy

The dataset is divided into training and testing sets using a 70/30 split. The split uses stratification, which means the percentage of churn and non-churn customers is preserved in both the training and testing sets. This is important because the dataset is imbalanced.

The implementation uses train_test_split with test_size=0.30, random_state=42, and stratify=y. The final test set contains 2,113 records, including 1,552 No Churn customers and 561 Churn customers. This confirms that the test set preserves the original churn distribution.

Split detail	Value
Training/testing split	70% training / 30% testing
Stratification	Used with stratify=y
Test set size	2,113 records
No Churn in test set	1,552 records
Churn in test set	561 records

5. Feature Analysis

Feature analysis was performed to understand which variables have the strongest relationship with churn. This stage also helps address the curse of dimensionality because one-hot encoding expands categorical variables into many columns. Without feature analysis, the model may include too many weak or redundant variables.

The project uses three approaches for feature analysis: correlation analysis, SelectKBest feature scoring, and Decision Tree feature importance. The exploratory charts below also support the same conclusions by showing how tenure, monthly charges, contract type, and payment method relate to churn behavior.

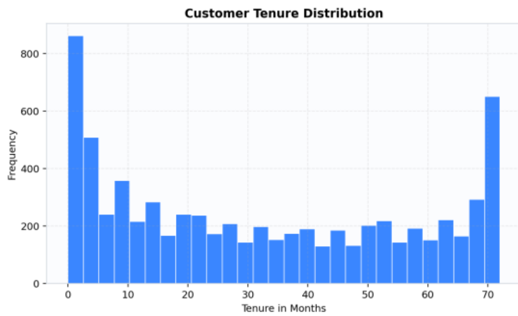

Figure 2. Tenure Distribution

Analysis: Tenure represents how long a customer has stayed with the company. Shorter-tenure customers are often more likely to leave.

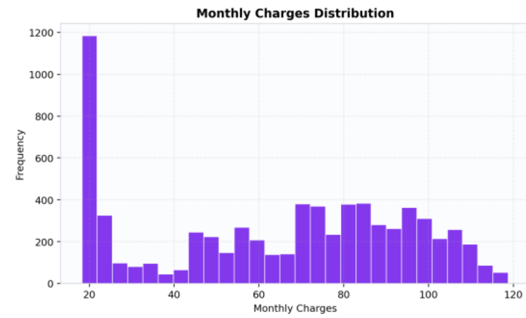

Figure 3. Monthly Charges Distribution

Analysis: Monthly charges vary widely. Higher monthly charges may influence churn when combined with short contracts or specific service types.

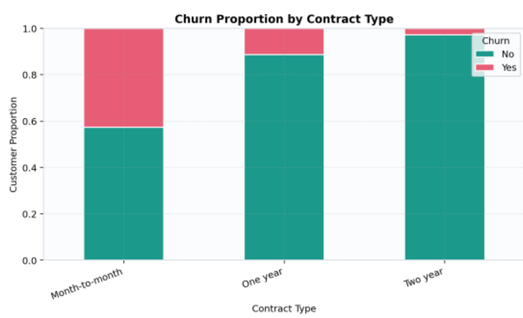

Figure 4. Contract Type vs Churn

Analysis: Month-to-month contracts are associated with higher churn, while two-year contracts are associated with lower churn.

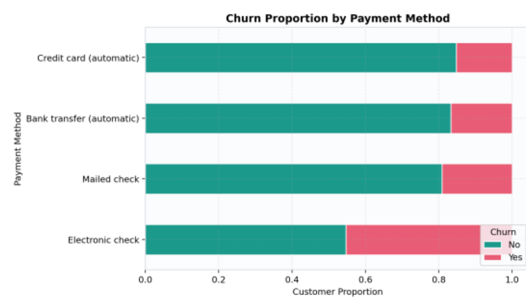

Figure 5. Payment Method vs Churn

Analysis: Electronic check payment is associated with higher churn compared with other payment methods.

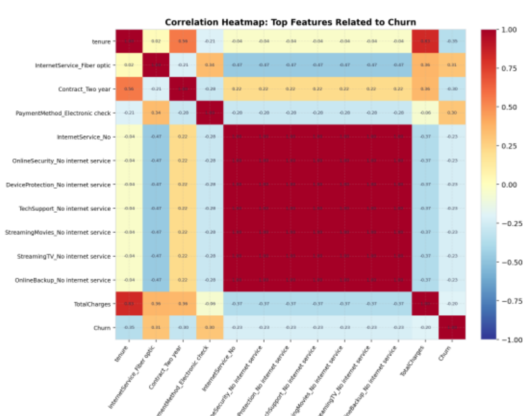

Figure 6. Correlation Matrix

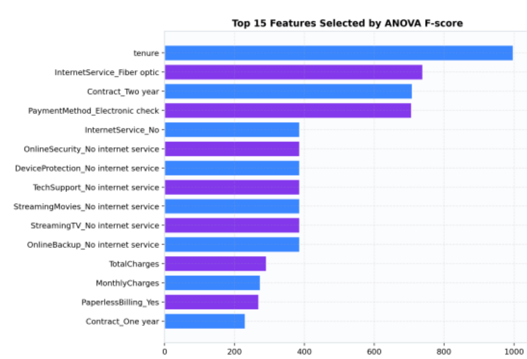

Figure 7. SelectKBest Top Features

Analysis: The strongest features include tenure, fiber optic internet, two-year

Analysis: Tenure and two-year contracts have a negative relationship with churn, while fiber optic internet and electronic check payment are positively associated with churn.

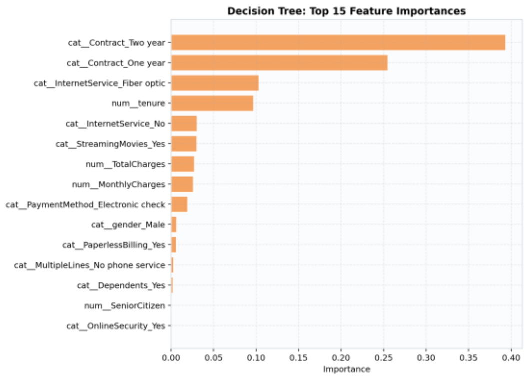

Figure 8. Decision Tree Feature Importance

Analysis: Decision Tree identifies contract type, internet service, tenure, charges, and payment method as important predictors.

contract, electronic check, TotalCharges, and MonthlyCharges.

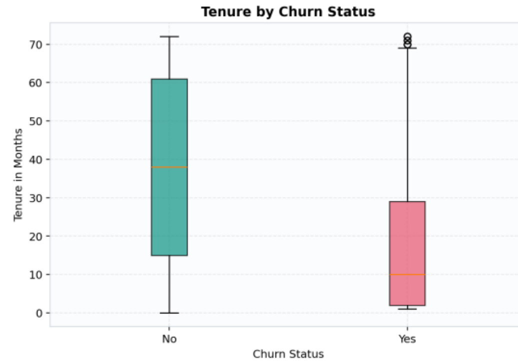

Figure 9. Tenure by Churn Boxplot

Analysis: The boxplot shows that churn customers tend to have lower tenure, confirming that new customers are more likely to leave.

Feature	SelectKBest Score	Selected
tenure	997.27	Yes
InternetService_Fiber optic	738.05	Yes
Contract_Two year	707.92	Yes
PaymentMethod_Electronic check	706.20	Yes
InternetService_No	385.70	Yes

6. Model Implementation

Two supervised classification algorithms were implemented and compared: Decision Tree and Support Vector Machine. Both models use class_weight="balanced" because the churn class is smaller than the no-churn class.

Model	Main parameters	Reason for use
Decision Tree Classifier	max_depth=5, random_state=42, class_weight="balanced"	Chosen because it is interpretable and provides feature importance and tree structure.
Support Vector Machine (SVM)	kernel="rbf", C=1.0, gamma="scale", class_weight="balanced", random_state=42	Chosen because it can model non-linear decision boundaries and often performs well on classification problems.

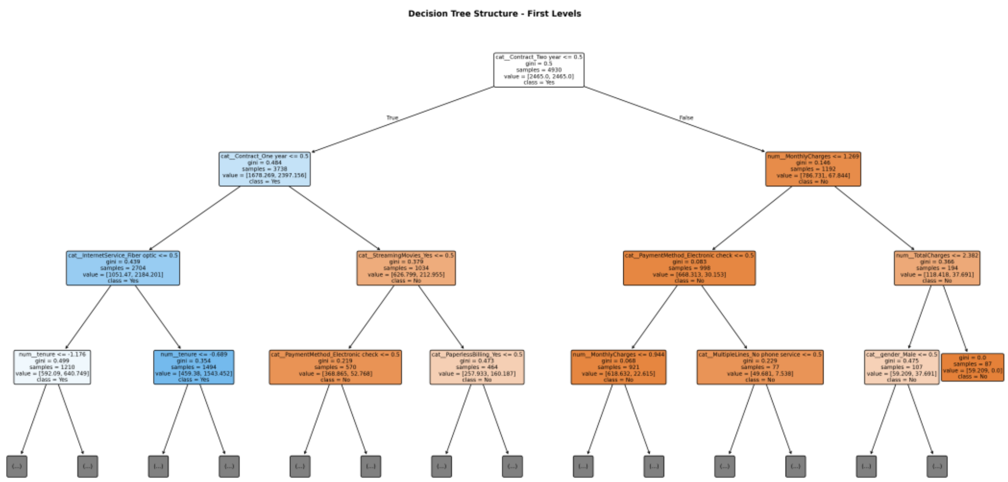

Figure 10. Decision Tree Structure

Analysis: The tree structure shows how the Decision Tree splits customers based on important variables. Limiting max_depth to 5 makes the model easier to interpret and reduces overfitting.

7. Experimental Results

The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Since the dataset is imbalanced, recall and F1-score are especially important. In churn prediction, recall is important because missing a real churn customer means the company may lose a customer without taking action.

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	71.56%	47.82%	78.25%	59.36%
SVM	74.54%	51.33%	78.97%	62.22%

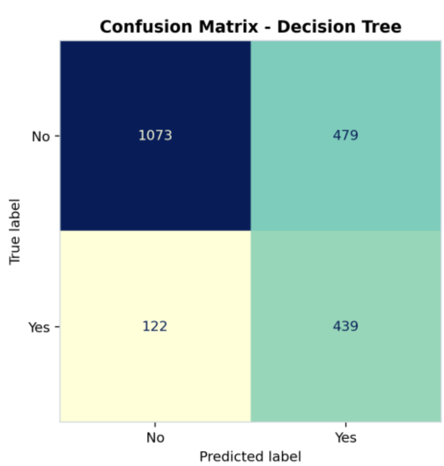

Figure 11. Decision Tree Confusion Matrix

Analysis: Decision Tree correctly identified 439 churn customers and missed 122 churn customers. It also classified 479 non-churn customers as churn, so recall is strong but precision is lower.

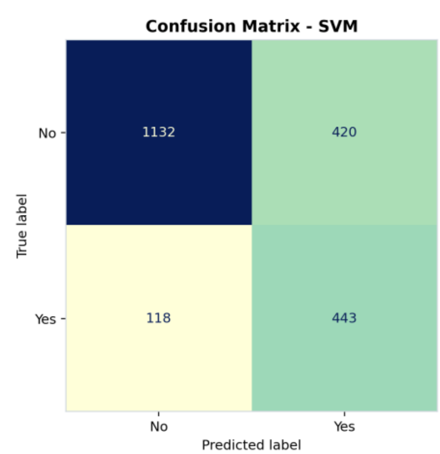

Figure 12. SVM Confusion Matrix

Analysis: SVM correctly identified 443 churn customers and missed 118 churn customers. It also reduced false positives to 420, which is better than Decision Tree.

Classification reports

The classification reports provide class-level precision, recall, and F1-score. For the churn class, SVM improves precision from 0.48 to 0.51 and F1-score from 0.59 to 0.62 while also slightly increasing recall.

Model	Class	Precision	Recall	F1-score	Support
Decision Tree	No Churn	0.90	0.69	0.78	1552
Decision Tree	Churn	0.48	0.78	0.59	561
SVM	No Churn	0.91	0.73	0.81	1552
SVM	Churn	0.51	0.79	0.62	561

8. Model Comparison

The comparison shows that SVM achieved the best overall performance. It has higher accuracy, precision, recall, and F1-score than the Decision Tree. However, the Decision Tree remains useful because it is easier to interpret and provides a clear feature importance ranking.

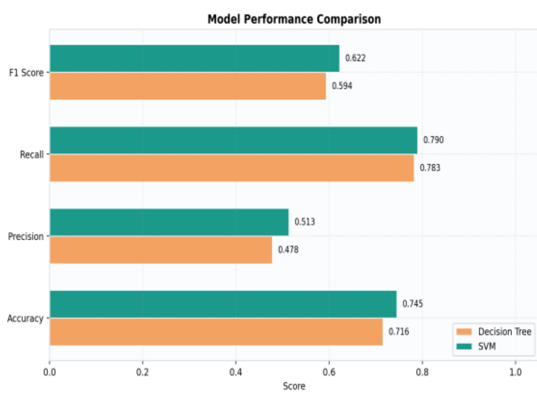

Figure 13. Model Comparison Chart

Analysis: SVM is higher in all four-evaluation metrics, confirming it is the stronger model for this dataset.

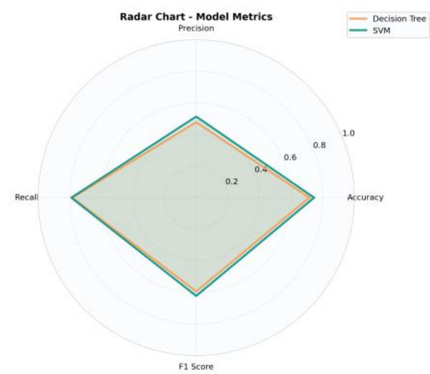

Figure 14. Radar Chart

Analysis: The SVM area is slightly larger than the Decision Tree area, giving a visual summary of better overall performance.

Criterion	Decision Tree	SVM	Better model
Accuracy	71.56%	74.54%	SVM
Precision	47.82%	51.33%	SVM
Recall	78.25%	78.97%	SVM
F1-score	59.36%	62.22%	SVM
Interpretability	High	Medium	Decision Tree

9. Discussion

The results provide both business and technical insights. Contract type, customer tenure, internet service type, payment method, and charges are the most influential factors for predicting churn. Customers with month-to-month contracts and electronic check payments are more likely to churn, while customers with longer tenure and two-year contracts are less likely to churn.

- Month-to-month customers are more likely to churn than customers with one-year or two-year contracts.
- Longer tenure is associated with lower churn, which means older customers are generally more loyal.
- Fiber optic internet service is strongly associated with churn in this dataset, possibly due to price, service expectations, or customer support issues.
- Electronic check payment is associated with higher churn compared with other payment methods.
- SVM performed better than Decision Tree in all main evaluation metrics, while Decision Tree remained more interpretable.

Additional K-Means clustering was also used to explore customer segments. It does not directly predict churn, but it helps discover groups of similar customers. The silhouette score was 0.2626, which indicates that the clusters exist but are not strongly separated. This is reasonable for real customer behavior data because customer patterns often overlap.

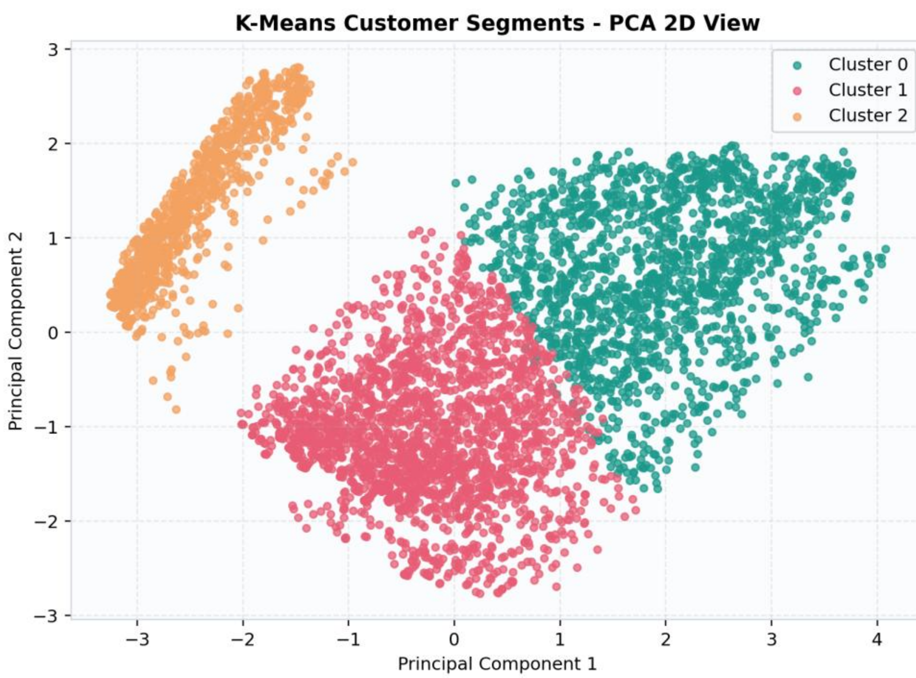

Figure 15. K-Means Clusters Visualized with PCA

Analysis: The PCA plot shows three customer segments. The moderate silhouette score indicates that the clusters are present but overlapping.

10. Conclusion

This project successfully applies a complete data mining and machine learning pipeline to the Telco Customer Churn dataset. The dataset was cleaned and transformed, categorical variables were encoded, numerical variables were scaled, and a stratified 70/30 train-test split was used. Feature analysis identified tenure, contract type, internet service, payment method, TotalCharges, and MonthlyCharges as important churn predictors.

Two required classification models were implemented and evaluated: Decision Tree and SVM. The SVM model achieved the best overall performance with 74.54% accuracy and 62.22% F1-score. The Decision Tree model was slightly weaker but more interpretable. Overall, the project meets the required report components: introduction, dataset description, data preprocessing, sampling strategy, feature analysis, model implementation, experimental results, model comparison, discussion, and conclusion.

The main recommendation is for the company to focus on customers with short tenure, month-to-month contracts, electronic check payments, and high monthly charges, since these groups appear more likely to churn.